

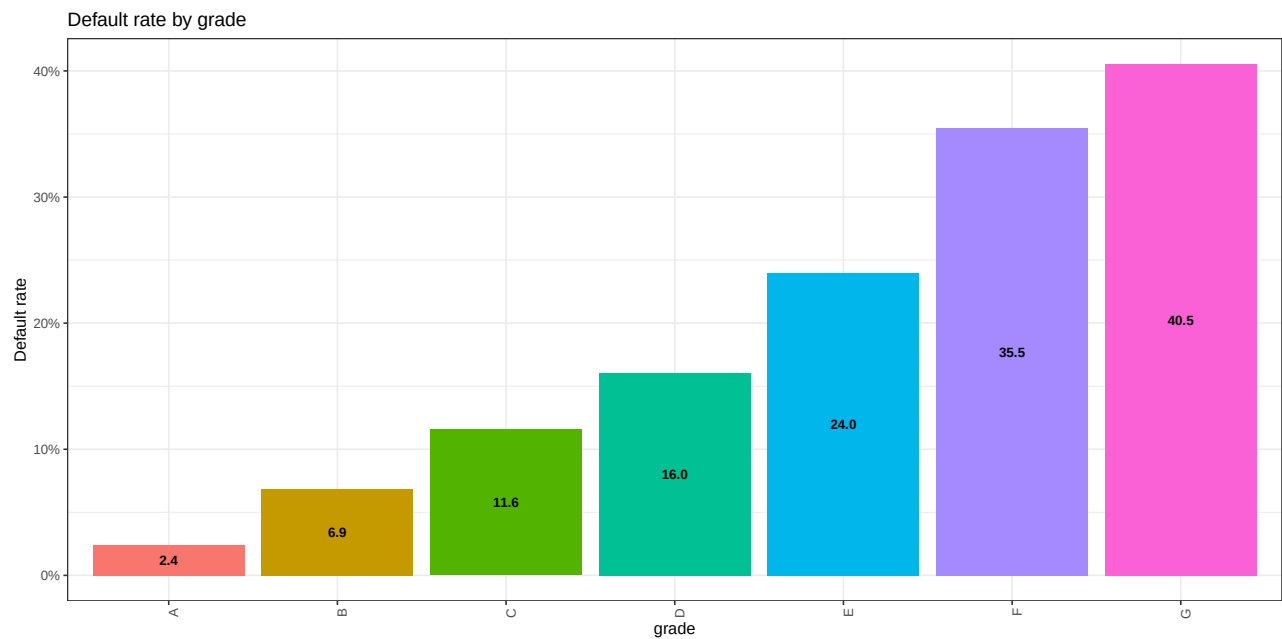
Loans and Credit

Anya Nash^a

^a*Stellenbosch University, Stellenbosch, South Africa*

0.1. Default Profile Summary Statistics

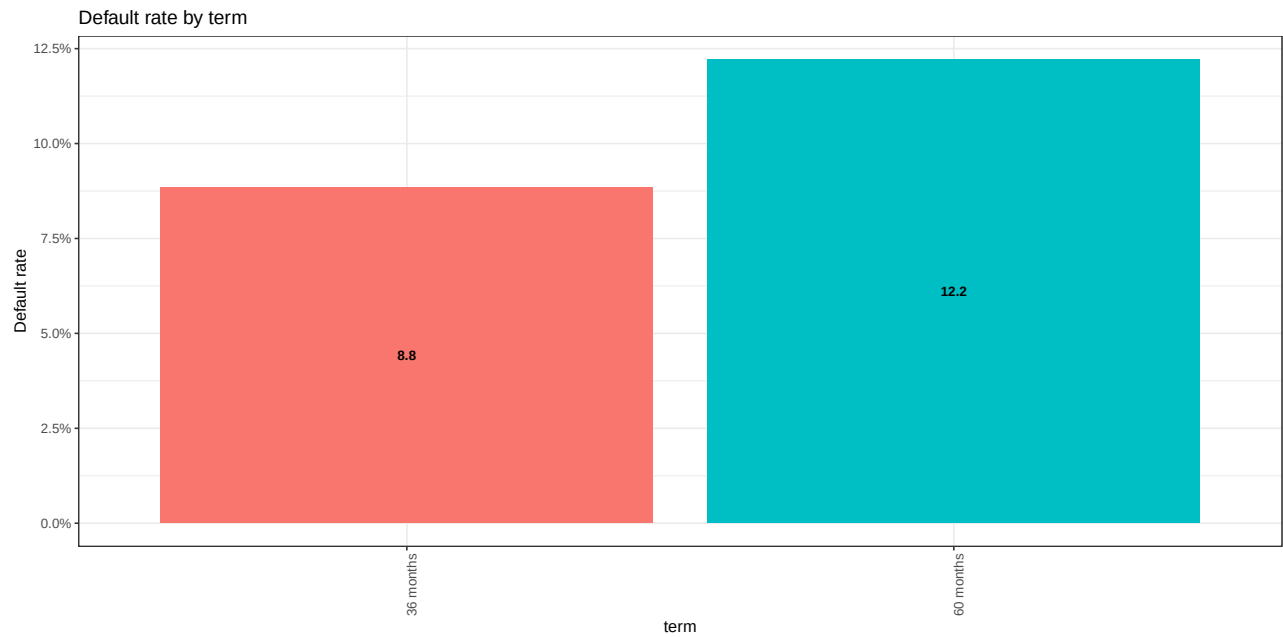
0.1.1. Default Rate by Grade



It is clear from the graph that as credit grade goes from A to G, the default rate increases exponentially. Therefore, grade is highly associated with the default rate (although that is how it is designed).

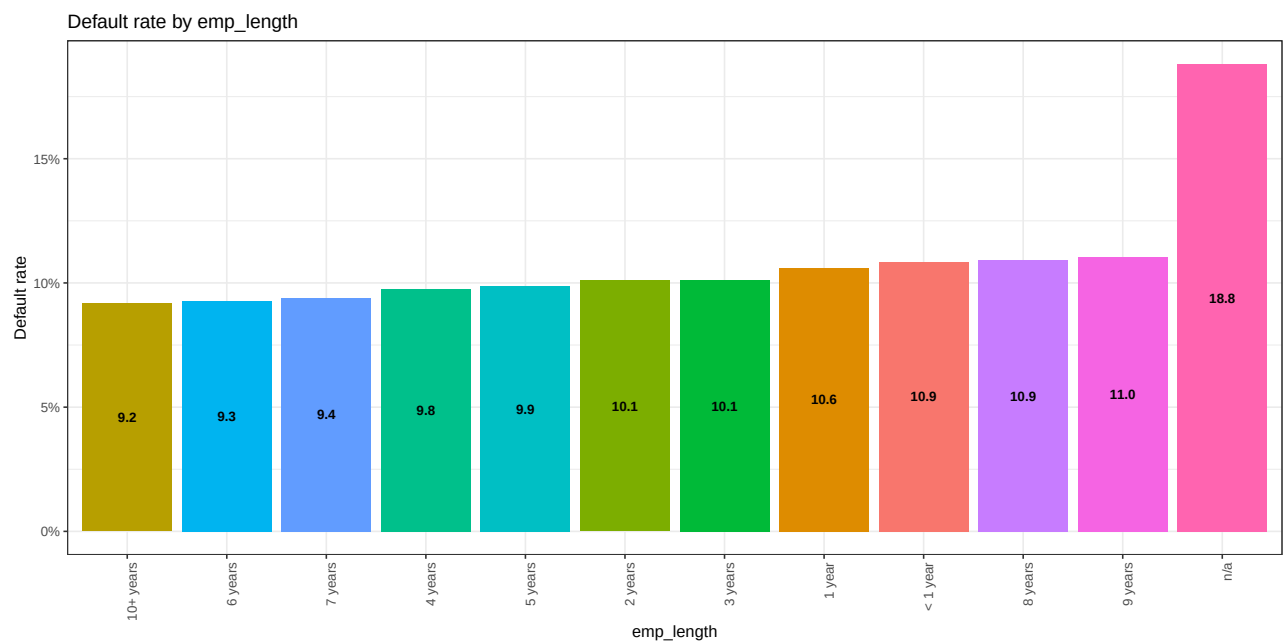
*Corresponding author: Anya Nash
Email address: 25051911@sun.ac.za (Anya Nash)

0.1.2. Default Rate by Term of Loan



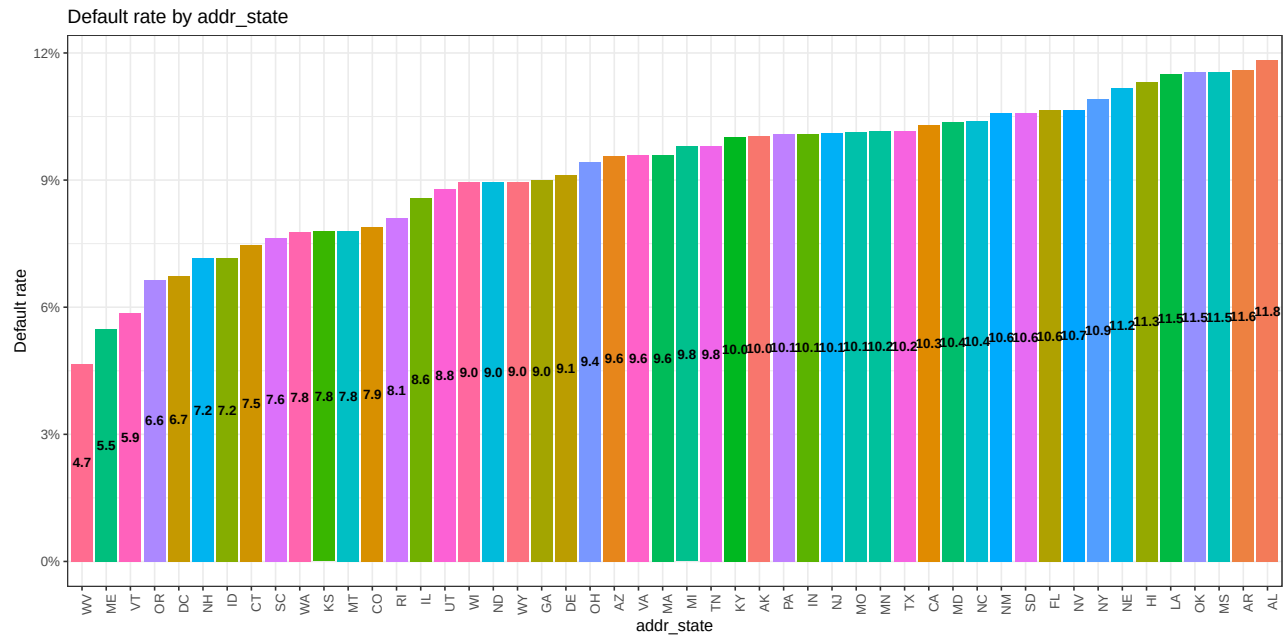
There is a small difference in the default rate based on the length of the loan, with the longer loan having a slightly higher default rate.

0.1.3. Default Rate by Length of Employment



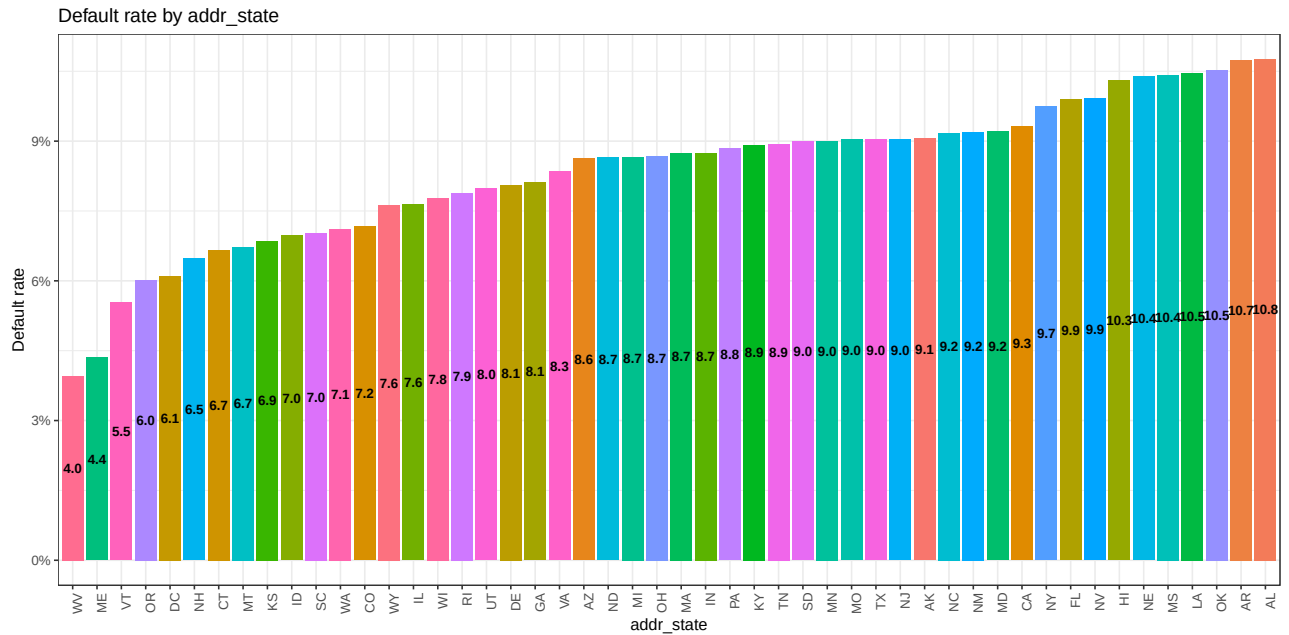
There is not a substantial difference in the default rate based on the length of employment, except for those who have n/a for length of employment - they have a much higher default rate. This is logical, because those who answered n/a to length of employment are most likely not working/ unemployed. Thus, they probably have a less stable income, which would make their default rate higher.

0.1.4. Default Rate by State



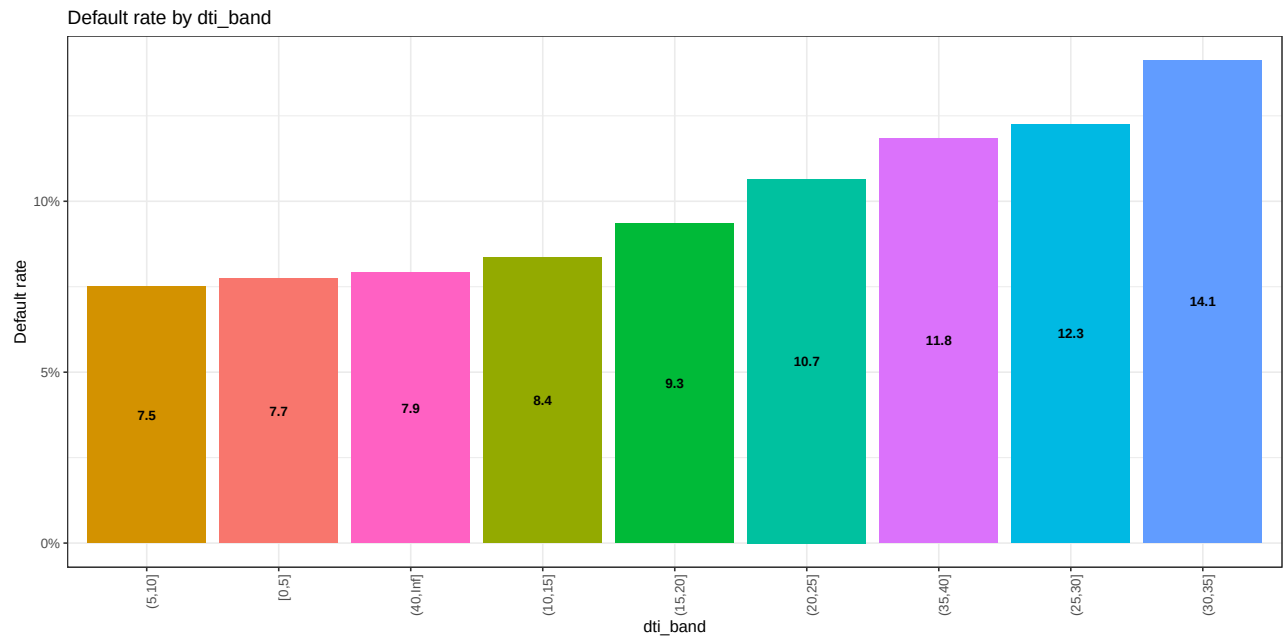
There is no obvious outlier in the default rate based on US state, however, the difference in default rate between the state with the highest default rate (Alabama) and the state with the lowest default rate (West Virginia) is substantial. Texas has a default rate on the higher side, however, it is not obviously different from any of the other states.

0.1.5. Default Rate by State for Short Term Loans



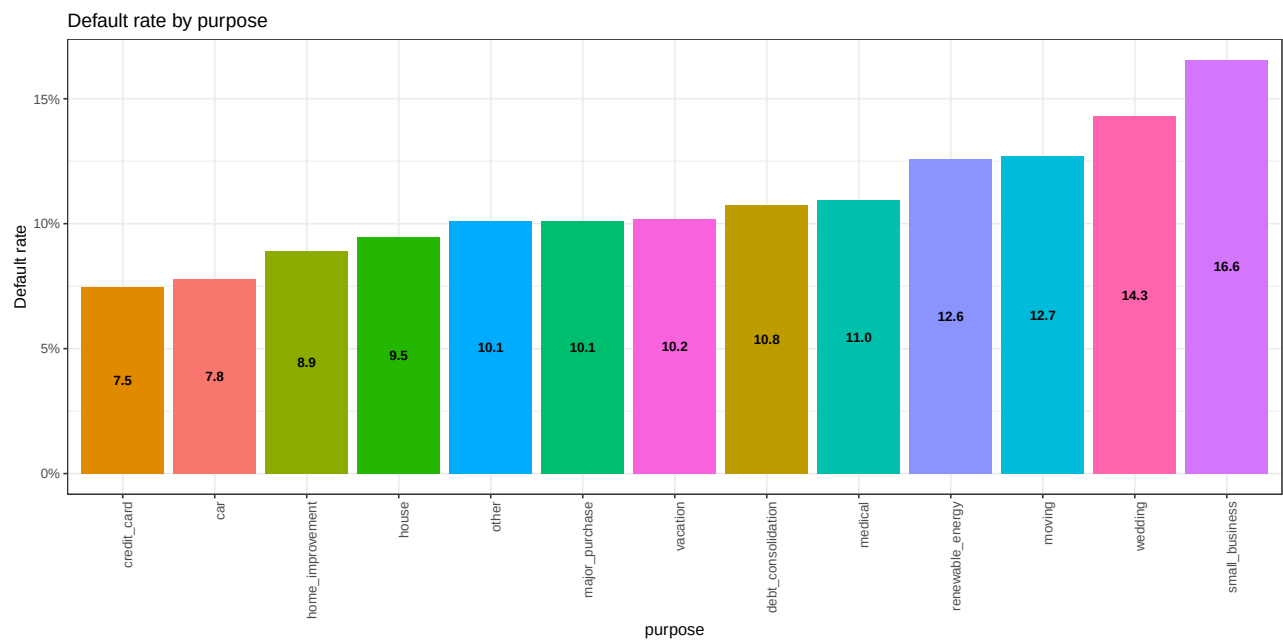
There is no obvious outlier in the default rate based on US state for short term loans, however, the difference in default rate between the state with the highest default rate (Alabama) and the state with the lowest default rate (West Virginia) is substantial. Texas has a default rate on the higher side, however, it is not obviously different from any of the other states.

0.1.6. Default Rate by DTI Band



The default rate for different bands of the debt-to-income ratio follows an expected pattern. The default rate is lower for a lower debt-to-income ratio, and the default rate is higher for a higher debt-to-income ratio.

0.1.7. Default Rate by Purpose of Loan



The default rate differs a bit based on the purpose of the loan. Loans for paying off credit cards or buying cars have the lowest default rate, while loans for weddings and small businesses have the highest default rate. It is logical that loans for small businesses have the highest default rate, since small businesses often carry a higher investment risk, and have unpredictable profits (and often low/no initial profits).

0.2. Default Rate, Credit Grades and Young Individuals

Unfortunately, there is no age variable in the data. Therefore, as a very rough, imperfect proxy for age, I use length of employment. I classify young individuals as those with a length of employment of 5 years or less. Then, to determine whether credit grades are a good indicator for whether younger individuals will default on their loans, I regress the default rate on credit grade, restricting the sample to younger individuals only. The table below shows the results of the regression.

Table 0.1: Regression of default rate on loan grade

Variable	Estimate	Std. Error	t-statistic	p-value
Constant	0.024	0.001	24.097	0
Grade B	0.048	0.001	36.969	0
Grade C	0.097	0.001	74.878	0
Grade D	0.141	0.002	90.907	0
Grade E	0.221	0.002	98.887	0
Grade F	0.331	0.004	84.018	0
Grade G	0.391	0.008	51.708	0

As seen in the regression table above, credit grade is a somewhat good indicator of whether younger individuals will default. This is because all the coefficients are statistically significant with an economically-meaningful magnitude, however, the R-squared value is quite low at 0.04928, meaning there are many other factors that influence the default rate of young individuals.

0.3. Interest Rates, Age, Occupation and Credit Scores

Using employment length as an imperfect, rough proxy for age once again and using very rough occupation categories, I regress interest rate on employment length, occupation and credit score to determine whether interest rates are determined by these factors. The results are displayed below.

```
## OLS estimation, Dep. Var.: int_rate
```

```
## Observations: 786,940
## Fixed-effects: grade: 7, emp_length: 12, emp_title_clean: 43,388
## RMSE: 1.3155 Adj. R2: 0.926009
```

With an R-squared value of 0.926009, it is clear that interest rates are strongly associated with age, occupation and credit scores.

0.4. Risk Profiles

The table below shows the top 10 combinations of credit grade, DTI band, purpose of loan and type of home ownership that have the highest default rates.

Table 0.2: Top 10 borrower profiles with the highest default rates

Grade	DTI band	Loan purpose	Home ownership	Number of borrowers	Default rate	Median annual income
G	(30,35]	debt_consolidation	RENT	202	53.5%	\$55,500
G	(25,30]	debt_consolidation	RENT	159	46.5%	\$56,000
F	(30,35]	debt_consolidation	RENT	597	44.7%	\$50,119
F	(20,25]	debt_consolidation	RENT	716	43.7%	\$60,000
G	(30,35]	debt_consolidation	MORTGAGE	186	41.9%	\$65,000
F	(25,30]	debt_consolidation	RENT	685	41.8%	\$55,000
G	(20,25]	debt_consolidation	RENT	172	40.7%	\$57,450
F	(35,40]	debt_consolidation	RENT	295	40.0%	\$50,000
G	(15,20]	debt_consolidation	RENT	118	39.8%	\$55,674
F	(15,20]	debt_consolidation	RENT	628	39.5%	\$62,000

From the table above, we can see that the highest defaulters are individuals with credit grades of either F or G, a DTI that is greater than 25 and who rent their homes. The median income for each group is between 50000 and 60000 US dollars.

When lenders assess borrower credibility, they can do a number of things to reduce default risk. They can identify those borrowers who are part of the highest default risk combination categories and either reduce the loan amount to these borrowers, or have additional checks and balances before approving the loan. Instead of just focusing on one category or “tick-box” at a time, lenders now have a better, fuller picture of the default risk of the borrower. A borderline answer for one or two categories could be viewed as okay when each category is viewed in isolation, but could paint a much riskier picture when categories are viewed together.

0.5. DTI Cap

The table below shows how the share of loans accepted and default rate would change, if there was a DTI cap imposed.

Table 0.3: Default risk under alternative DTI hard caps

DTI cap	Borrowers accepted	Share accepted	Default rate
5	47688	5.3%	7.7%
10	164287	18.1%	7.6%
15	341407	37.7%	8.0%
20	528833	58.4%	8.5%
25	684950	75.6%	9.0%
30	798324	88.1%	9.4%
35	864625	95.4%	9.8%
40	891675	98.4%	9.9%
45	896435	98.9%	9.9%
50	899763	99.3%	9.9%
55	901461	99.5%	9.9%

As seen in the table above, as the DTI cap increases, both the share accepted and default rate increase. Therefore, there is a trade-off between the risk of default and maximising the number of borrowers. A DTI cap that is too strict will cause the number of borrowers to decline significantly, however, it would also have the positive effect of reducing the default rate. The same goes for a DTI cap that is too lenient - the number of borrowers will increase, but at the cost of the default rate increasing as well. I think an appropriate hard-cap for DTI levels is 25. At 25, approximately 75% of borrowers are accepted, which is not too low, and the default rate is at 9%, which is an improvement compared to no DTI cap.